

CSE330: Numerical Methods

Assignment 4

10 Marks

1. A function is given by $f(x) = e^{0.5x} + \sin(x)$. Given limit for integration $[0, 2]$

- a. (1 marks) Evaluate the exact integral $I(f)$.
- b. (3 marks) Compute the numerical integral by using the Newton-Cotes formula with $n=1$.
- c. (3 marks) Compute the numerical integral by using the Newton-Cotes formula With Simpsons rule.
- d. (3 marks) Compute the numerical integral by using the Composite Newton-Cotes formula for 4 sub divisions.

Bonus Question:

For the scenario above find the error bound or highest possible value for $|I(f) - I_n(f)|$

Hint: See the lecture note